MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, SWITCHES,

MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Quad bilateral switch
02	Quad bilateral switch
51	Quad bilateral switch
52	Quad bilateral switch

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	F-1 (14-lead, 1/4" x 1/4"), flat package, except A dimension = 0.1" (2.54 mm maximum)
Y	F-2 (14-lead, 1/4" x 3/8"), flat package, except A dimension = 0.1" (2.54 mm maximum)

NOTES:
1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outlines X and Y only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-1 and F-2) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.

2. For bottom or side braded packages, case outlines X and Y only, the S_1 dimension may go to .000 inch (.00 mm) mimimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA Parts Project Office, Code 311.A, NASA/Goddard Space Flight Center, Greenbelt, MD 20771 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range (V_{DD}-V_{SS})	
Device types 01 and 02	-0.5 V to +15.5 V
Device types 51 and 52	-0.5 V to +18.0 V
Input current (each input)	±10 mA
Input voltage range	$(V_{SS} - 0.5\ V) \leq V_I \leq (V_{DD} + 0.5\ V)$
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D)	175 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC})	(See MIL-M-38510, appendix C)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage:	
Device types 01 and 02	4.5 V dc to 12.5 V dc
Device types 51 and 52	4.5 V dc to 15 V dc
Minimum high-level input voltage (V_{IH}):	
Device types 01 and 02	2.05 - 5.0 V dc @ V_{DD} = 5 V; 11.85 - 12.5 V dc @ V_{DD} = 12.5 V
Device types 51 and 52	3.5 - 5.0 V dc at V_{DD} = 5 V, 8.0 - 10.0 V dc @ V_{DD} = 10 V, 11.0 - 15.0 V dc @ V_D = 15 V V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}
Minimum low-level input voltage range (V_{IL}):	
Device types 01 and 02	0 - 0.95 V dc @ V_{DD} = 5 V; 0 - 2.4 V dc @ V_{DD} = 12.5 V
Device types 51 and 52	0 - 1.5 V dc at V_{DD} = 5 V, 0 - 2.0 V dc @ V_{DD} = 10 V, 0 - 4.0 V dc @ V_{DD} = 15 V, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}
Load capacitance	50 pF maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. *REQUIREMENTS*

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 *Conductive Silver Paste, or equivalent which is cured at* 200°C ±10°C *for a minimum* of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 1.

3.2.2 Truth table. The truth table shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits, including protection networks, shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with *MIL-M-38510 and 6.4 herein.*

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 39 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in initial (pre-burn-in) electrical parameters (3.1.10) through interim (post-burn-in) electrical parameters (3.1.14) and substitute lines 1 through 7 of table II herein.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/ V_{SS} = 0 V $-55°C \leq T_C = \leq 125°C$ Unless otherwise specified	Device type	Limits		Unit
				Min	Max	
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_C = 25°C, V_{DD} = GND V_{SS} = Open, Output = Open, I_I = 1 mA	All	---	1.5	V
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_C = 25°C, V_{DD} = Open V_{SS} = GND, Output = Open I_I = -1 mA	All	---	-6	V
Quiescent supply current	I_{SS}	V_{DD} = 15 V dc any combination of inputs	All	---	-550	nA
High level output voltage	V_{OH1}	V_{IN} = 5 V, V_{DD} = 5 V dc, I_{OH} = 0, Control input = V_{IH} (see table III)	01, 02	4.95	---	V
	V_{OH2}	V_{IN} = 12.5 V, V_{DD} = 12.5 V dc, I_{OH} = 0, Control input = V_{IH} (see table III)	01, 02	11.25	---	V
	V_{OH3}	V_{IN} = 15 V, V_{DD} = 15 V dc I_{OH} = 0, Control input = V_{IH}	51, 52	14.95	---	V
Low level output voltage	V_{OL1}	V_{IN} = 0 V, V_{DD} = 5 V dc, I_{OL} = 0, All control inputs = V_{IH} (see table III)	01, 02	---	-.05	V
	V_{OL2}	V_{IN} = 0 V, V_{DD} = 12.5 V dc, I_{OL} = 0, All control inputs = V_{IH} (see table III)	01, 02	---	1.25	V
	V_{OL3}	V_{IN} = 0 V, V_{DD} = 15 V dc, I_{OL} = 0, All control inputs = V_{IN}	51, 52	---	0.05	V
Input high control voltage	V_{IH1}	V_{DD} = 5 V dc V_O = 0.5 V $\lvert I_O \rvert \leq 1$ µA	51, 52	3.5	---	V
	V_{IH2}	V_{DD} = 10 V dc V_O = 1.0 V $\lvert I_O \rvert \leq 1$ µA	51, 52	7.0	---	V
	V_{IH3}	V_{DD} = 15 V dc V_O = 1.5 V dc $\lvert I_O \rvert \leq 1$ µA	51, 52	11.0	---	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_C = ≤ 125°C, Unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Input low control voltage	V_{IL1}	V_{DD} = 5 V dc, V_O ≤ 0.5 V $\|I_O\|$ ≤ 1 µA	52	---	1.5	V
			51	---	1.0	V
	V_{IL2}	V_{DD} = 10 V dc, V_O ≤ 1.0 V $\|I_O\|$ ≤ 1 µA	52	---	3.0	V
			51	---	2.0	V
	V_{IL3}	V_{DD} = 15 V dc, V_O ≤ 1.5 V dc $\|I_O\|$ ≤ 1 µA	52	---	4.0	V
			51	---	2.5	V
Output low (sink) current	I_{OL1}	V_{DD} = 5 V dc, V_{IN} = 5 V, V_{OL} = 0.4 V dc	51, 52	0.36	---	mA dc
	I_{OL2}	V_{DD} = 15 V dc, V_{IN} = 15 V, V_{OL} = 1.5 V dc	51, 52	2.4	---	mA dc
Output high (source) current	I_{OH1}	V_{DD} = 5 V dc, V_{OH} = 4.6 V, V_{IN} = 5 V	51, 52	-0.36	---	mA dc
	I_{OH2}	V_{DD} = 15 V dc, V_{IN} = 15 V, V_{OH} = 13.5 V dc	51, 52	-2.4	---	mA dc
Input leakage current 2/	I_{IH}	Measure inputs sequentially V_{DD} = 15 V dc, V_{DD} = 18 V dc	01, 02, 51, 52	---	45	nA
	I_{IL}	Measure inputs sequentially V_{DD} = 15 V dc, V_{DD} = 18 V dc	01, 02, 51, 52	---	-45	nA
Input test voltage	V_{ZAP}	C_L = 100 pF R_2 = 1.5 kΩ (see 4.5.3)	All	400	---	V
Input capacitance	C_{ic}	V_{DD} = 0 V dc, f = 1 MHz, T_C = 25°C	All	---	12	pF
Input cap	C_{is}	V_{DD} = 5 V dc, f = 1 MHz, T_C = 25°C	01, 51		15	pF
			02, 52		20	
Output cap	C_{os}		01, 51		15	pF
			02, 52		20	
Feedthrough cap	C_{ics}		01, 51		15	pF
			02, 52		20	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/ V_{SS} = 0 V, -55°C ≤ T_C ≤ 125°C, Unless otherwise specified	Device type	Limits Min	Limits Max	Unit
Propagation delay times, switch input to output	t_{PHL}	V_{DD} = 5 V dc, C_L = 50 pF (see figure 5), R_L = 200 kΩ	01, 51	4	156	ns
			02, 52	4	80	ns
	t_{PLH}		01, 51	4	130	ns
			02, 52	4	85	ns
Propagation delay times, control input to switch output	t_{PZH}, t_{PZL}, t_{PHZ}, t_{PLZ}	V_{DD} = 5 V dc, C_L = 50 pF (see figure 6), R_L = 1 kΩ	All	4	90	ns
Switch "ON" resistance	R_{ON1}	V_{DD} = 5.0 V dc, V_{SS} = GND	01, 51	---	1.8k	Ω
			02, 52	---	600	Ω
	R_{ON2}	V_{DD} = 7.5 V dc, V_{SS} = -7.5 V dc	01, 51	---	450	Ω
			02, 52	---	250	Ω
	R_{ON3}	V_{DD} = 10.0 V dc, V_{SS} = GND	01, 51	---	750	Ω
			02, 52	---	300	Ω
Switch leakage current	I_{DOFF1}	V_{DD} = +5.0 V dc, V_{SS} = 0 V dc	All	---	45	nA
	I_{DOFF2}	V_{DD} = 0 V dc, V_{SS} = -5.0 V dc				
	I_{DOFF3}	V_{DD} = +10.0 V dc, V_{SS} = 0 V dc				
	I_{DOFF4}	V_{DD} = 0 V dc, V_{SS} = -10.0 V dc				
	I_{DOFF5}	V_{DD} = +15 V dc, V_{SS} = 0 V dc				
	I_{DOFF6}	V_{DD} = 0 V dc, V_{SS} = -15.0 V dc				

1/ Complete terminal conditions shall be as specified in table III.
2/ Input current at node.

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S device 1/				Class B device 1/			
		Ref. par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III sub-groups	Table 1/ IV delta limits	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2			3				
3	Same as line 1		1	Δ					
4	Static burn in II (method 1015)	4.2b 4.5.2			3	4.2d 4.5.2	4/		3
5	Same as line 1	4.2d	1*	Δ		4.2	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2			4				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters (method 5004)		1*,2,3,9				1*,2,3,9		
9	Group A end-point electrical parameters (method 5005)	4.4.1	1,2,3,4, 9,10,11			4.4.1	1,2,3, 4,9,10, 11		
10	Group B end-point electrical parameters (method 5005)	4.4.2c	1,2,3,9, 10,11	*					
11	Group C end-point electrical parameters (method 5005)					4.4.3	1,2,3	Δ	
12	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2,3		

1/ Blank spaces indicate tests are not applicable.

2/ (*) indicates PDA applies to subgroup 1 (see 4.2.1).

3/ (Δ) indicates delta limit shall be required on table III subgroup 1 where specified, and delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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b. Burn-in (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 3 or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 4 or equivalent. Ambient temperture (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The PDA for class S devices shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed to burn-in lots. The verified failures divided by the total number of devices in lots initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D and E inspections (see 4.4.1 through 4.4.5).

4.3.1 Qualification extension. When authorized by the qualifying activity, if a manufacturer qualifies to a 51 and 52 device type which is manufactured identically to a 01 and 02 device type on this specification, then the 01 and 02 device type may be part I qualified by conducting only group A electrical and submitting data in accordance with MIL-M-38510, appendix D.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5, 6, 7, and 8 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroups 9 and 11 shall be measured only for initial qualification and after process or design changes which may affect dynamic performance.

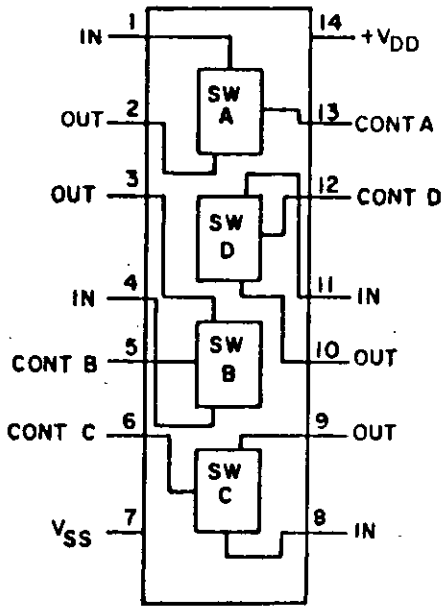
e. At the manufacturer's option, test tapes may be programmed simultaneously for each identical section provided that each output is measured and each specified input combination is tested.

f. When device types 01 and 02 are qualified by extension (see 4.3.1) they shall be inspected (QCI) in accordance with the requirements for corresponding device types 51 and 52.

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Device types 01 and 51

Cases A, C, D, X, and Y



Device types 02 and 52

Cases A, C, D, X, and Y

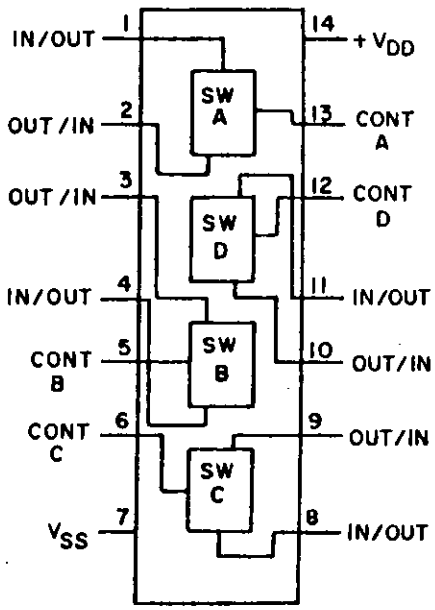


FIGURE 1. Terminal connections.

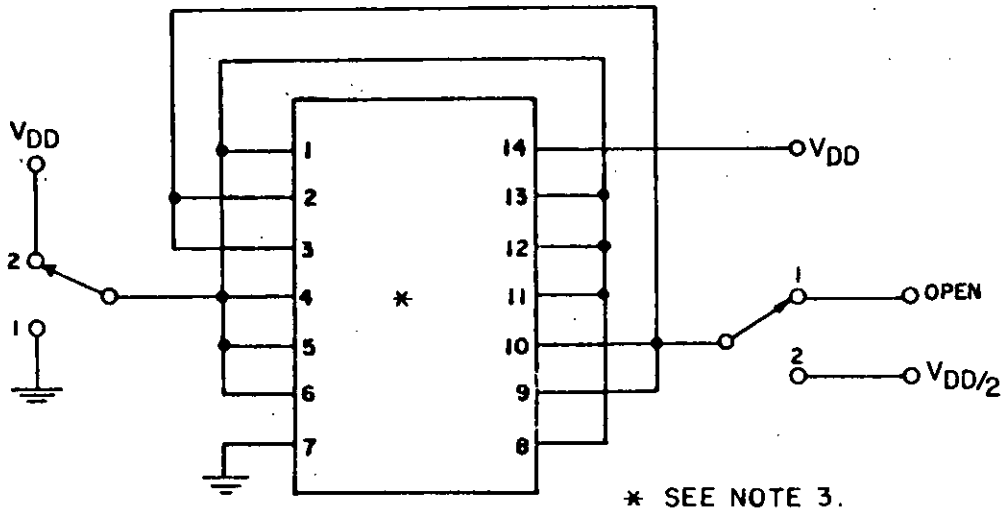
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Truth table each switch		
Input		Output
V_C	V_{IS}	V_{OS}
1	0	0
1	1	1
0	0	Open
0	1	Open

Positive logic: Switch on V_C = "1"
Switch off V_C = "0"

FIGURE 2. Truth table.

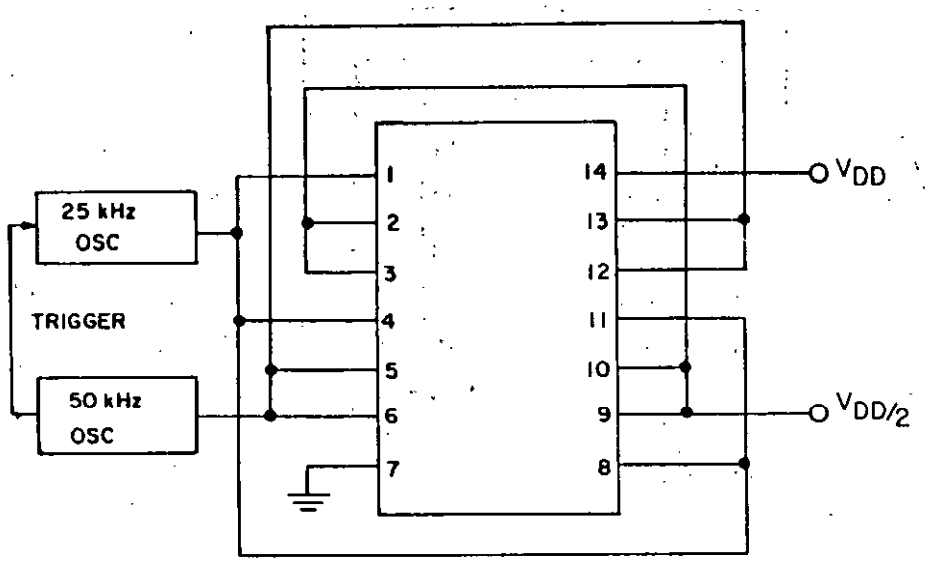
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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, terminals shall be connected through a resistor(s) whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.
5. V_{DD} = 12.5 V minimum, 15 V maximum for device types 01, and 02.
   V_{DD} = 15 V minimum, 18 V maximum for device types 51, and 52.
   $V_{DD/2}$ = $V_{DD/2}$ ±1.0 V for all devices.
   V_{SS} = 0 V.

FIGURE 3. Static burn-in test circuits.

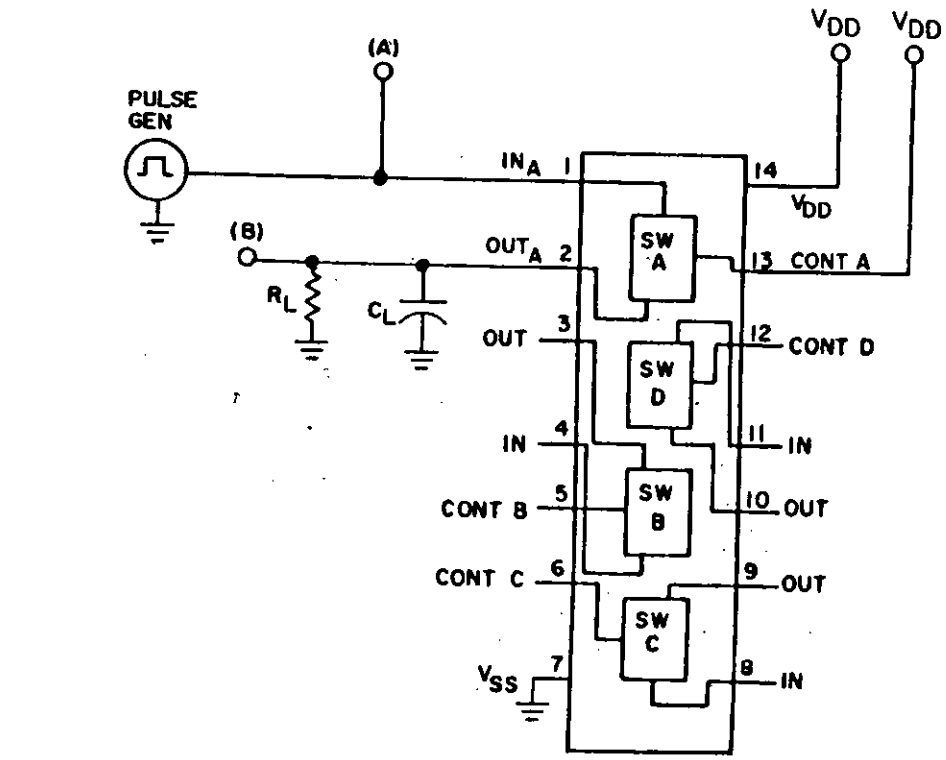
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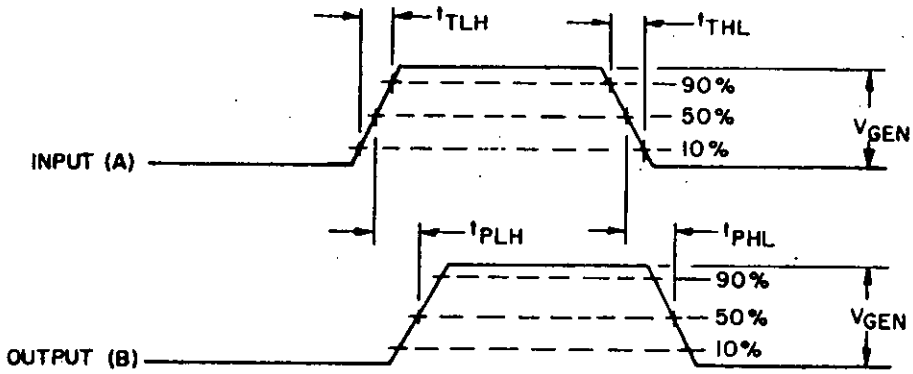


NOTES:
1. Except for V_{DD} and V_{SS}, terminals shall be connected through a resistor(s) whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
3. V_{DD} = 12.5 V minimum, 15.0 V maximum for device types 01, and 02.
   V_{DD} = 15 V minimum, 18 V maximum for device types 51, and 52.
   $V_{DD/2}$ = $V_{DD/2}$ ±1.0 V.
   V_{SS} = 0 V.

FIGURE 4. Dynamic burn-in and steady state life test circuit.

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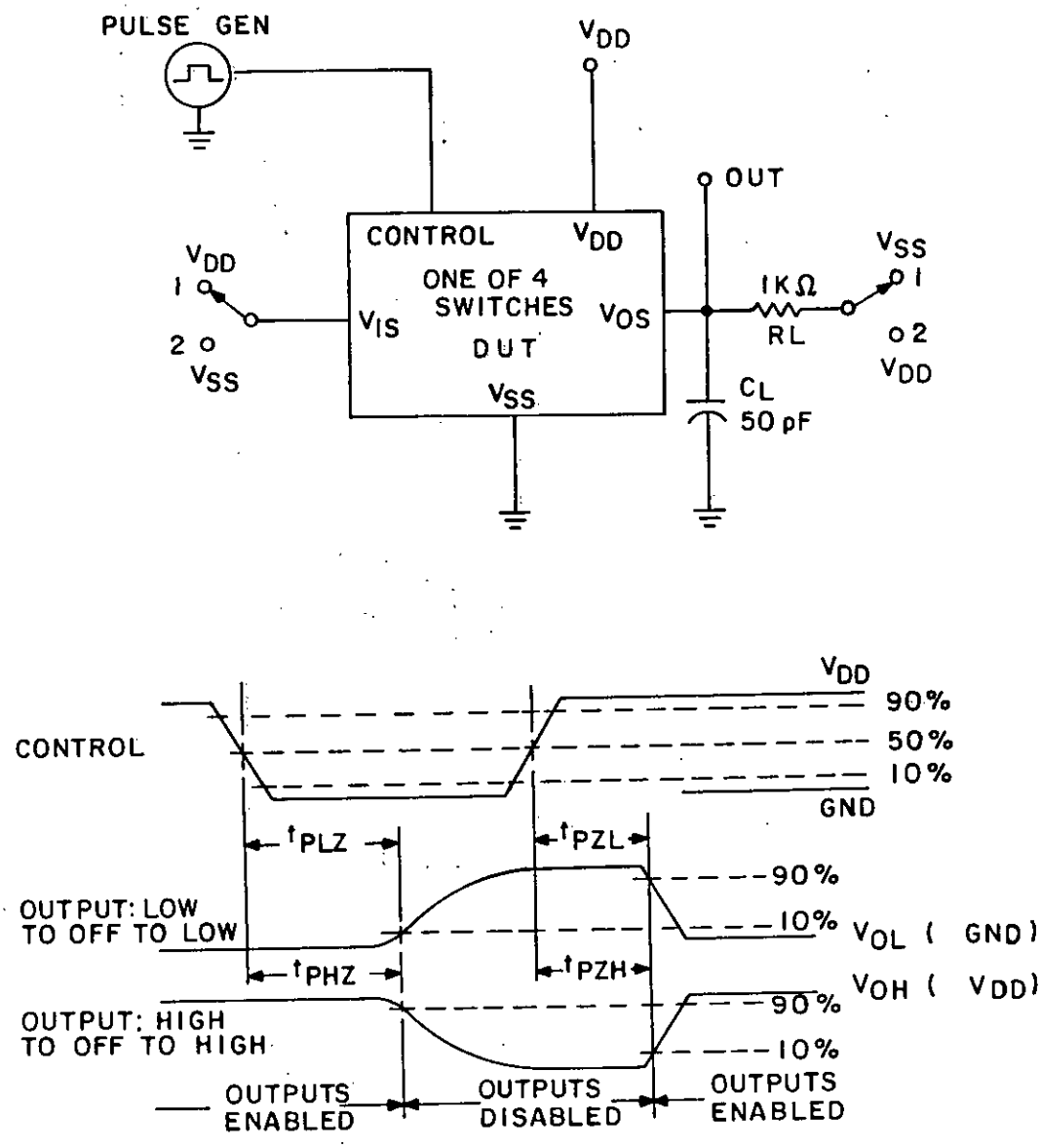




NOTES:
1. Pulse generator conditions: $V_{GEN} = V_{DD}$ ±1%, duty cycle = 50%, t_{THL} and t_{TLH} = 20 ns ±2.0 ns, PRR = 500 KHz, and Z_{OUT} = 50Ω.
2. C_L = 50 pF, R_L = 200 K ±10% ohms, includes probe and jig impedance.
3. Identical switching measurements are obtained from switch A, switch B, switch C and switch D.

FIGURE 5. Switching time test circuit and waveforms, switch input to switch output.

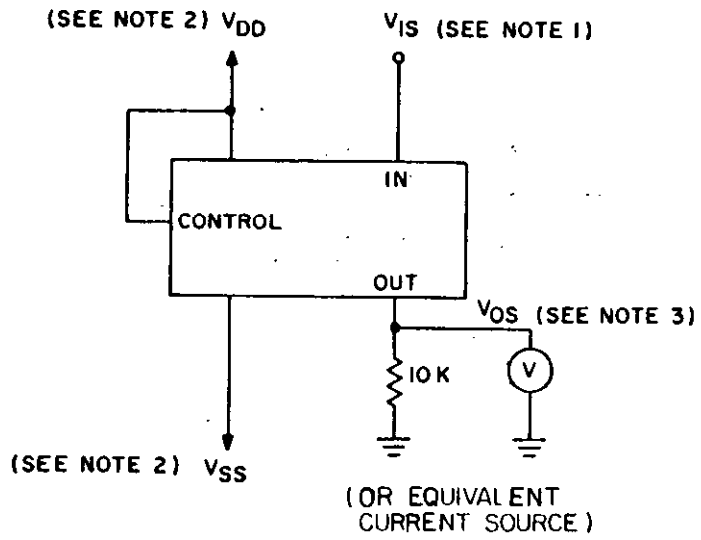
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:07Z
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NOTES:
1. Pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, duty cycle = 50%, t_r and $t_f \leq 20$ ns and $Z_{OUT} = 50\Omega$.
2. $C_L = 50$ pF, $R_L = 1k\Omega$, includes probe and jig impedance.
3. Identical switching measurements are obtained from switch A, switch B, switch C and switch D.
4. Switch pos. 1 for t_{pzH}, t_{pzH}, t_{pHz}, test; switch pos. 2 for t_{pLz}, t_{pzL}.

FIGURE 6. Switching time test circuit and waveforms, switch control input to switch output.

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NOTES:

1.

R_{ON1} NUMBER	R_{ON1} (1)	R_{ON1} (2)	R_{ON1} (3)	R_{ON1} (4)
V_{IS}	+0.5 V	1.0 V	4.0 V	5.0 V

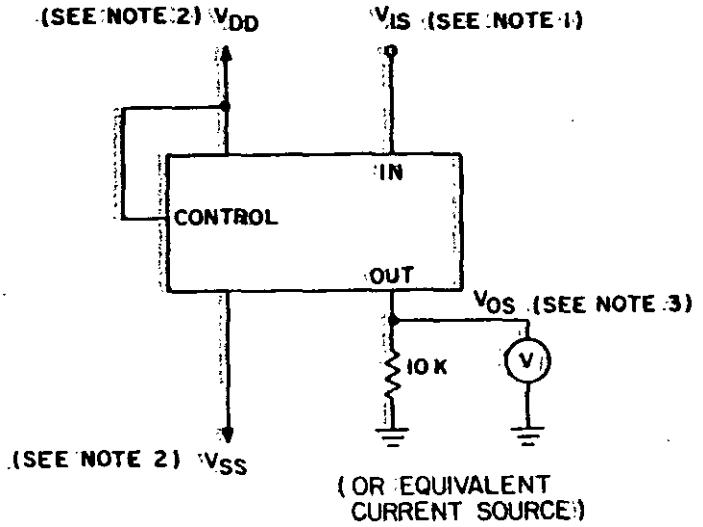
2. Conditions V_{DD} and V_{SS}: V_{DD} = +5.0 V, V_{SS} = 0 V.
3. Translation from V_{OS} to R_{ON}:

$$R_{ON} = \frac{V_{IS} - V_{OS}}{V_{OS} / 10\ K\Omega}$$

4. See table III for complete terminal conditions.

FIGURE 7. ON resistance test.

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NOTES:

1.

R_{ON2} NUMBER	R_{ON2} (1)	R_{ON2} (2)	R_{ON2} (3)
V_{IS}	-5.0 V	+0.6 V	+5.0 V

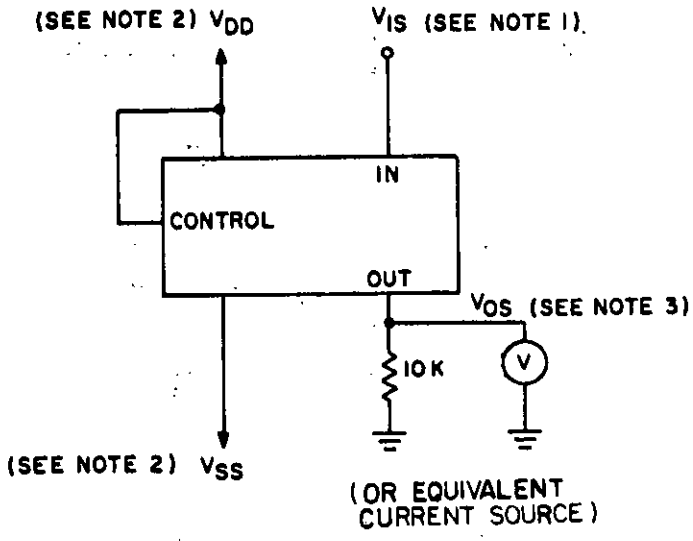
2. Conditions V_{DD} and V_{SS}: V_{DD} = +7.5 V, V_{SS} = -7.5 V.
3. Translation from V_{OS} to R_{ON}:

$$R_{ON} = \frac{V_{IS} - V_{OS}}{V_{OS} / 10\ K\Omega}$$

4. See table III for complete terminal conditions.
5. At the manufacturer's option, during R_{ON2} (1) tests, V_{DD}, V_{SS}, and V_{IS} may be varied provided the absolute value equals +15.0 V.

FIGURE 8. ON resistance test.

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NOTES:

1.

R_{ON3} NUMBER	R_{ON3} (1)	R_{ON3} (2)	R_{ON3} (3)
V_{IS}	+2.5 V	5.0 V	7.5 V

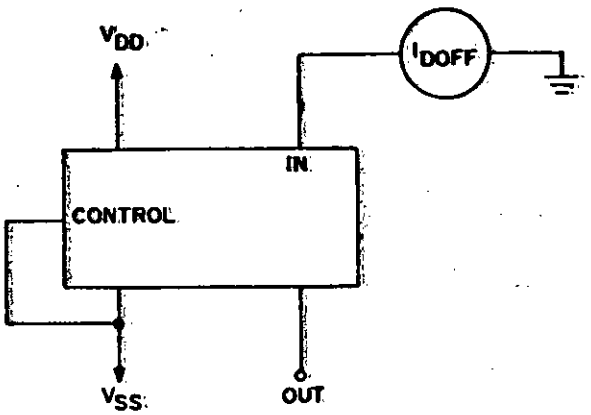
2. Conditions V_{DD} and V_{SS}: V_{DD} = +10.0 V, V_{SS} = 0 V.
3. Translation from V_{OS} to R_{ON}:

$$R_{ON} = \frac{V_{IS} - V_{OS}}{V_{OS} / 10\ K\Omega}$$

4. See table III for complete terminal conditions.

FIGURE 9. ON resistance test.

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NOTES:

1.

I_{DOFF} NUMBERS	V_{DD}	V_{SS}	Out
I_{DOFF-1}	+5.0 V	GND	+2.5 V
I_{DOFF-2}	GND	-5.0 V	-2.5 V
I_{DOFF-3}	+10.0 V	GND	+5.0 V
I_{DOFF-4}	GND	-10.0 V	-5.0 V
I_{DOFF-5}	+15.0 V	GND	+7.5 V
I_{DOFF-6}	GND	-15.0 V	-7.5 V

2. Identical measurements shall be performed on switch A, switch B, switch C and switch D.

FIGURE 10. Switch leakage current.

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TABLE III. Group A inspection for device type 01. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits						Unit
																		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1 2 3 4					1 mA	1 mA						1 mA	1 mA	GND " "	C_A C_B C_C C_D		1.5 " "					V " "
$V_{IC(neg)}$		5 6 7 8					-1 mA	-1 mA	GND " "					-1 mA	-1 mA		C_A C_B C_C C_D		-6 " "					" " "
I_{SS} 2/ 3/	3005 "	9 10 11	15 V GND "	GND 15 V	GND 15 V	15 V GND "	GND " 15 V	GND " "	" " "	15 V GND "	GND 15 V	GND 15 V	15 V GND "	GND " 15 V	GND " 15 V	15 V " "	V_{SS} "		-30 " "		-550 " "			nA "
V_{OH1} 4/	3006 " "	12 13 14 15	 GND 5.0 V			5.0 V GND	V_{IH1} V_{IL1} GND "	V_{IL1} V_{IH1} GND "	" " " "	" 5.0 V			 5.0 V GND	GND " V_{IH1} V_{IL1}	GND " V_{IL1} V_{IH1}	5.0 V " " "	O_B O_C O_D O_A	4.95 " "		4.95 " "		4.95 " "		V " " "
V_{OH2} 4/	" " "	16 17 18 19	 GND 12.5 V			12.5 V GND	V_{IH2} V_{IL2} GND "	V_{IL2} V_{IH2} GND "	" " " "	GND 12.5 V			 12.5 V GND	GND " V_{IH2} V_{IL2}	GND " V_{IL2} V_{IH2}	12.5 V " " "	O_B O_C O_D O_A	11.25 " "		11.25 " "		11.25 " "		" " " "
V_{OL1} 4/	3007 " "	20 21 22 23	 5.0 V GND			GND 5.0 V	V_{IH1} V_{IL1} GND "	V_{IL1} V_{IH1} GND "	" " " "	5.0 V GND			 GND 5.0 V	GND " V_{IH1} V_{IL1}	GND " V_{IL1} V_{IH1}	5.0 V " " "	O_B O_C O_D O_A		50 " "		50 " "		50 " "	mV " "
V_{OL2} 4/	" " "	24 25 26 27	 12.5 V GND			GND 12.5 V	V_{IH2} V_{IL2} GND "	V_{IL2} V_{IH2} GND "	" " " "	5.0 V GND			 GND 12.5 V	GND " V_{IH2} V_{IL2}	GND " V_{IL2} V_{IH2}	12.5 V " " "	O_B O_C O_D O_A		1.25 " "		1.25 " "		1.25 " "	V " "
I_{IH1} 5/	3010	28	15 V			15 V	15 V	15 V	"	15 V			15 V	15 V	15 V	15 V	All inputs together		+12					nA
I_{IH2}	" " " " " "	29 30 31 32 33 34 35 36	GND " " " 15 V GND " "			GND " " " " 15 V GND "	15 V GND " " " " " "	GND 15 V GND " " " " "	" " " " " " " "	GND " " " " " 15 V GND			GND " " " " " " 15 V	GND " 15 V GND " " " "	GND " 15 V GND " " " "	" " " " " " " "	C_B C_C C_D C_A I_A I_B I_C I_D		1.0 " " 2.0 " " "		45 " " " " " "			" " " " " " "
I_{IL1} 5/	3009	37	"			"	"	"	"	"			GND	"	"	"	All inputs together		-12					"

See footnotes at end of device type 02.

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TABLE III. Group A inspecttion for device type 01 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C,D,X,Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 1 T_C = 25°C Min	Subgroup 1 T_C = 25°C Max	Subgroup 2 T_C = 125°C Min	Subgroup 2 T_C = 125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
I_{IL2}	3009	38	GND			GND	GND	GND	GND	GND			GND	GND	GND	15 V	C_B		-1.0		-45			nA
	"	39	"			"	"	"	"	"			"	"	"	"	C_C		"		"			"
	"	40	"			"	"	"	"	"			"	"	"	"	C_D		"		"			"
	"	41	"			"	"	"	"	"			"	"	"	"	C_A		"		"			"
	"	42	"			"	"	"	"	"			"	"	"	"	I_A		-2.0		"			"
	"	43	"			"	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	44	"			"	"	"	"	"			"	"	"	"	I_C		"		"			"
	"	45	"			"	"	"	"	"			"	"	"	"	I_D		"		"			"
R_{ON1}		46	0.5 V	6/											5.0 V	5.0 V	O_A		700		750		550	Ω
		47															O_B		"		"		"	"
		48						5.0 V		0.5 V	6/						O_C		"		"		"	"
		49			6/							6/	0.5 V	5.0 V			O_D		"		"		"	"
		50	1.0 V	6/											5.0 V		O_A		1.0 k		1.1 k		750	"
		51				1.0 V	5.0 V										O_B		"		"		"	"
		52						5.0 V		1.0 V	6/						O_C		"		"		"	"
		53			6/							6/	1.0 V	5.0 V			O_D		"		"		"	"
		54	4.0 V	6/											5.0 V		O_A		1.2 k		1.8 k		850	"
		55			6/	4.0 V	5.0 V										O_B		"		"		"	"
		56						5.0 V		4.0 V	6/						O_C		"		"		"	"
		57										6/	4.0 V	5.0 V			O_D		"		"		"	"
		58	5.0 V	6/											5.0 V		O_A		800		950		750	"
		59			6/	5.0 V	5.0 V										O_B		"		"		"	"
		60						5.0 V		5.0 V	6/						O_C		"		"		"	"
		61										6/	5.0 V	5.0 V			O_D		"		"		"	"
R_{ON2}		62	-5.0 V	7/					-7.5 V						7.5 V	7.5 V	O_A		250		300		250	Ω
		63							"								O_B		"		"		"	"
		64			7/			7.5 V		-5.0 V	7/						O_C		"		"		"	"
		65				-5.0 V	7.5 V					7/	-5.0 V	7.5 V			O_D		"		"		"	"
		66	0.6 V	7/											7.5 V		O_A		400		450		300	"
		67			7/	0.6 V	7.5 V										O_B		"		"		"	"
		68						7.5 V		0.6 V	7/						O_C		"		"		"	"
		69										7/	0.6 V	7.5 V			O_D		"		"		"	"
		70	5.0 V	7/											7.5 V		O_A		350		400			"
		71			7/	5.0 V	7.5 V										O_B		"		"			"
		72						7.5 V		5.0 V	7/						O_C		"		"			"
		73										7/	5.0 V	7.5 V			O_D		"		"			"
R_{ON3}		74	2.5 V	8/					GND						10 V	10 V	O_A		450		500		350	Ω
		75							"								O_B		"		"		"	"
		76			8/	2.5 V	10 V										O_C		"		"		"	"
		77						10 V		2.5 V	8/						O_D		"		"		"	"
		78	5.0 V	8/								8/	2.5 V	10 V			O_A		650		700		500	"
		79			8/	5.0 V	10 V								10 V		O_B		"		"		"	"
		80						10 V		5.0 V	8/						O_C		"		"		"	"
		81										8/	5.0 V	10 V			O_D		"		"		"	"
		82	7.5 V	8/											10 V		O_A		600		750			"
		83			8/	7.5 V	10 V										O_B		"		"			"
		84						10 V		7.5 V	8/						O_C		"		"			"
		85										8/	7.5 V	10 V			O_D		"		"			"
I_{DOFF1}	Fig. 10	86	9/	9/	9/	9/	9/	9/	"	9/	9/	9/	9/	9/	9/	5.0 V	9/		45		45		45	nA
I_{DOFF2}	"	87	"	"	"	"	"	"	-5.0 V	"	"	"	"	"	"	GND	"		"		"		"	"
I_{DOFF3}	"	88	"	"	"	"	"	"	GND	"	"	"	"	"	"	10 V	"		"		"		"	"
I_{DOFF4}	"	89	"	"	"	"	"	"	-10 V	"	"	"	"	"	"	GND	"		"		"		"	"
I_{DOFF5}	"	90	"	"	"	"	"	"	GND	"	"	"	"	"	"	15 V	"		"		"		"	"
I_{DOFF6}	"	91	"	"	"	"	"	"	-15 V	"	"	"	"	"	"	GND	"		"		"		"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 01 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits: Subgroup 4 T_C = 25°C Min	Subgroup 4 T_C = 25°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}				
C_C	3012	92					10/		GND							GND	C_B		12	pF
	"	93						10/	"					10/		"	C_C		"	"
	"	94							"						10/	"	C_D		"	"
	"	95							"							"	C_A		"	"
C_{is}	3012	96	10/				GND	GND	GND					GND	GND	5.0 V	I_A		15	"
	"	97				10/	"	"	"					"	"	"	I_B		"	"
	"	98					"	"	"	10/				"	"	"	I_C		"	"
	"	99					"	"	"				10/	"	"	"	I_D		"	"
C_{OS}		100		11/			"	"	"					"	"	"	O_A		"	"
		101			11/		"	"	"		11/			"	"	"	O_B		"	"
		102					"	"	"			11/		"	"	"	O_C		"	"
		103					"	"	"					"	"	"	O_D		"	"
C_{ios}		104	12/	12/			"	"	"					"	"	"	I_A, O_A		"	"
		105			12/	12/	"	"	"	12/	12/			"	"	"	I_B, O_B		"	"
		106					"	"	"			12/	12/	"	"	"	I_C, O_C		"	"
		107					"	"	"					"	"	"	I_D, O_D		"	"

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Subgroup 9 T_C = 25°C Min	Subgroup 9 T_C = 25°C Max	Subgroup 10 T_C = 125°C Min	Subgroup 10 T_C = 125°C Max	Subgroup 11 T_C = -55°C Min	Subgroup 11 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
t_{PLH}	3003 Fig. 5	108	IN	OUT	OUT	IN	GND	GND	GND	IN	OUT	OUT	IN	GND	5.0 V GND	5.0 V	I_A to O_A	4	125	7	130	4	114	ns
	"	109					5.0 V	5.0 V	"					"	"	"	I_B to O_B	"	"	"	"	"	"	"
	"	110					GND	GND	"					5.0 V	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	111					"	"	"					"	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHL}	"	112	IN	OUT	OUT	IN	"	"	"	IN	OUT	OUT	IN	GND	5.0 V GND	"	I_A to O_A	"	130	"	156	"	130	"
	"	113					5.0 V	5.0 V	"					"	"	"	I_B to O_B	"	"	"	"	"	"	"
	"	114					GND	GND	"					5.0 V	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	115					"	"	"					"	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHZ}	Fig. 6	116	5.0 V	OUT	OUT	5.0 V	"	"	"	5.0 V	OUT	OUT	5.0 V	GND	IN GND	"	C_A to O_A	"	70	"	90	"	50	"
	"	117					IN	IN	"					"	"	"	C_B to O_B	"	"	"	"	"	"	"
	"	118					GND	GND	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	119					"	"	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZH}	Fig. 6	120	5.0 V	OUT	OUT	5.0 V	"	"	"	5.0 V	OUT	OUT	5.0 V	GND	IN GND	"	C_A to O_A	"	"	"	"	"	"	"
	"	121					IN	IN	"					"	"	"	C_B to O_B	"	"	"	"	"	"	"
	"	122					GND	GND	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	123					"	"	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PLZ}	Fig. 6	124	GND	OUT	OUT	GND	"	"	"	GND	OUT	OUT	GND	GND	IN GND	"	C_A to O_A	"	"	"	"	"	"	"
	"	125					IN	IN	"					"	"	"	C_B to O_B	"	"	"	"	"	"	"
	"	126					GND	GND	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	127					"	"	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZL}	Fig. 6	128	GND	OUT	OUT	GND	"	"	"	GND	OUT	OUT	GND	GND	IN GND	"	C_A to O_A	"	"	"	"	"	"	"
	"	129					IN	IN	"					"	"	"	C_B to O_B	"	"	"	"	"	"	"
	"	130					GND	GND	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	131					"	"	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 1 T_C = 25°C Min	Subgroup 1 T_C = 25°C Max	Subgroup 2 T_C = 125°C Min	Subgroup 2 T_C = 125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
$V_{IC(pos)}$		1					1 mA								1 mA	GND	C_B		1.5					V
		2						1 mA								"	C_C		"					"
		3														"	C_D		"					"
		4												1 mA		"	C_A		"					"
$V_{IC(neg)}$		5							GND						-1 mA		C_B		-6					"
		6					-1 mA		"								C_C		"					"
		7						-1 mA	"								C_D		"					"
		8							"					-1 mA			C_A		"					"
I_{SS} 2/3/	3005	9	15 V	GND	GND	15 V	GND	GND	"	15 V	GND	GND	15 V	GND	GND	15 V	V_{SS}		-30		-550			nA
	"	10	GND	15 V	15 V	GND	"	"	"	GND	15 V	15 V	GND	"	"	"	"		"		"			"
	"	11	"			"	15 V	"	"	"			"	15 V	15 V	"								
V_{OH1} 4/	3006	12				5.0 V	V_{IH1}	V_{IL1}	"	"				GND	GND	5.0 V	O_B	4.95		4.95		4.95		V
	"	13				GND	V_{IL1}	V_{IH1}	"	5.0 V				"	"	"	O_C	"		"		"		"
	"	14	GND				GND	GND	"				5.0 V	V_{IH1}	V_{IL1}	"	O_D	"		"		"		"
	"	15	5.0 V				"	"	"				GND	V_{IL1}	V_{IH1}	"	O_A	"		"		"		"
V_{OH2} 4/	"	16				12.5 V	V_{IH2}	V_{IL2}	"	GND				GND	GND	12.5 V	O_B	11.25		11.25		11.25		"
	"	17				GND	V_{IL2}	V_{IH2}	"	12.5 V				"	"	"	O_C	"		"		"		"
	"	18	GND				GND	GND	"				12.5 V	V_{IH2}	V_{IL2}	"	O_D	"		"		"		"
	"	19	12.5 V				"	"	"				GND	V_{IL2}	V_{IH2}	"	O_A	"		"		"		"
V_{OL1} 4/	3007	20				GND	V_{IH1}	V_{IL1}	"	5.0 V				GND	GND	5.0 V	O_B		50		50		50	mV
	"	21				5.0 V	V_{IL1}	V_{IH1}	"	GND				"	"	"	O_C		"		"		"	"
	"	22	5.0 V				GND	GND	"				GND	V_{IH1}	V_{IL1}	"	O_D		"		"		"	"
	"	23	GND				"	"	"				5.0 V	V_{IL1}	V_{IH1}	"	O_A		"		"		"	"
V_{OL2} 4/	"	24				GND	V_{IH2}	V_{IL2}	"	5.0 V				GND	GND	12.5 V	O_B		1.25		1.25		1.25	V
	"	25					V_{IL2}	V_{IH2}	"	GND				"	"	"	O_C		"		"		"	"
	"	26					GND	GND	"				GND	V_{IH2}	V_{IL2}	"	O_D		"		"		"	"
	"	27					"	"	"				12.5 V	V_{IL2}	V_{IH2}	"	O_A		"		"		"	"
I_{IH1} 5/	3010	28	15 V			15 V	15 V	15 V	"	15 V			15 V	15 V	15 V	15 V	All inputs together		+12					nA
I_{IH2}	"	29	GND			GND	15 V	GND	"	GND			GND	GND	GND	"	C_B		1.0		45			"
	"	30	"			"	GND	15 V	"	"			"	"	"	"	C_C		"		"			"
	"	31	"			"	"	GND	"	"			"	15 V	"	"	C_D		"		"			"
	"	32	"			"	"	"	"	"			"	GND	15 V	"	C_A		"		"			"
	"	33	15 V			"	"	"	"	"			"	"	GND	"	I_A		2.0		"			"
	"	34	GND			15 V	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	35	"			GND	"	"	"	15 V			"	"	"	"	I_C		"		"			"
	"	36	"			"	"	"	"	GND			15 V	"	"	"	I_D		"		"			"
I_{IL1} 5/	3009	37	"			"	"	"	"	"			GND	"	"	"	All inputs together		-12					"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits						Unit
																		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}		Min	Max	Min	Max	Min	Max	
I_{IL2}	3009	38	GND			GND	GND	GND	GND	GND			GND	GND	GND	15 V	C_B		-1.0		-45			nA
	"	39	"			"	"	"	"	"			"	"	"	"	C_C		"		"			"
	"	40	"			"	"	"	"	"			"	"	"	"	C_D		"		"			"
	"	41	"			"	"	"	"	"			"	"	"	"	C_A		"		"			"
	"	42	"			"	"	"	"	"			"	"	"	"	I_A		-2.0		"			"
	"	43	"			"	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	44	"			"	"	"	"	"			"	"	"	"	I_C		"		"			"
	"	45	"			"	"	"	"	"			"	"	"	"	I_D		"		"			"
R_{ON1}		46	0.5 V	6/											5.0 V	5.0 V	O_A		400		450		400	Ω
		47			6/	0.5 V	5.0 V		"	0.5 V	6/						O_B		"		"		"	"
		48						5.0 V	"			6/	0.5 V	5.0 V			O_C		"		"		"	"
		49							"						5.0 V	"	O_D		"		"		"	"
		50	1.0 V	6/					"								O_A		450		500		450	"
		51			6/	1.0 V	5.0 V		"	1.0 V	6/						O_B		"		"		"	"
		52						5.0 V	"			6/	1.0 V	5.0 V			O_C		"		"		"	"
		53							"						5.0 V	"	O_D		"		"		"	"
		54	4.0 V	6/					"								O_A		550		600		550	"
		55			6/	4.0 V	5.0 V		"	4.0 V	6/						O_B		"		"		"	"
		56						5.0 V	"			6/	4.0 V	5.0 V			O_C		"		"		"	"
		57							"						5.0 V	"	O_D		"		"		"	"
		58	5.0 V	6/					"								O_A		400		450		400	"
		59			6/	5.0 V	5.0 V		"	5.0 V	6/						O_B		"		"		"	"
		60						5.0 V	"			6/	5.0 V	5.0 V			O_C		"		"		"	"
		61							"								O_D		"		"		"	"
R_{ON2}		62	-5.0 V	7/					-7.5 V						7.5 V	7.5 V	O_A		200		250		200	"
		63			7/	-5.0 V	7.5 V		"	-5.0 V	7/						O_B		"		"		"	"
		64						7.5 V	"			7/	-5.0 V	7.5 V			O_C		"		"		"	"
		65							"						7.5 V	"	O_D		"		"		"	"
		66	0.6 V	7/					"								O_A		250		300		250	"
		67			7/	0.6 V	7.5 V		"	0.6 V	7/						O_B		"		"		"	"
		68						7.5 V	"			7/	0.6 V	7.5 V			O_C		"		"		"	"
		69							"						7.5 V	"	O_D		"		"		"	"
		70	5.0 V	7/					"								O_A		300		350		300	"
		71			7/	5.0 V	7.5 V		"	5.0 V	7/						O_B		"		"		"	"
		72						7.5 V	"			7/	5.0 V	7.5 V			O_C		"		"		"	"
		73							"						7.5 V	"	O_D		"		"		"	"
R_{ON3}		74	2.5 V	8/					GND						10 V	10 V	O_A		300		350		300	"
		75			8/	2.5 V	10 V		"	2.5 V	8/						O_B		"		"		"	"
		76						10 V	"			8/	2.5 V	10 V			O_C		"		"		"	"
		77							"						10 V	"	O_D		"		"		"	"
		78	5.0 V	8/					"								O_A		250		300		250	"
		79			8/	5.0 V	10 V		"	5.0 V	8/						O_B		"		"		"	"
		80						10 V	"			8/	5.0 V	10 V			O_C		"		"		"	"
		81							"						10 V	"	O_D		"		"		"	"
		82	7.5 V	8/					"								O_A		"		"		"	"
		83			8/	7.5 V	10 V		"	7.5 V	8/						O_B		"		"		"	"
		84						10 V	"			8/	7.5 V	10 V			O_C		"		"		"	"
		85							"								O_D		"		"		"	"
I_{DOFF1}	Fig. 10	86	9/	9/	9/	9/	9/	9/	"	9/	9/	9/	9/	9/	9/	5 V	9/		45		45		45	nA
I_{DOFF2}	"	87	"	"	"	"	"	"	-5.0 V	"	"	"	"	"	"	GND	"		"		"		"	"
I_{DOFF3}	"	88	"	"	"	"	"	"	GND	"	"	"	"	"	"	10 V	"		"		"		"	"
I_{DOFF4}	"	89	"	"	"	"	"	"	-10 V	"	"	"	"	"	"	GND	"		"		"		"	"
I_{DOFF5}	"	90	"	"	"	"	"	"	GND	"	"	"	"	"	"	15 V	"		"		"		"	"
I_{DOFF6}	"	91	"	"	"	"	"	"	-15 V	"	"	"	"	"	"	GND	"		"		"		"	"

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 02 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C,D,X,Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits Subgroup 4 T_C = 25°C Min	Subgroup 4 Max	Min	Max	Min	Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
C_C	3012	92					10/		GND							GND	C_B		12					pF
	"	93						10/	"					10/		"	C_C		"					"
	"	94							"						10/	"	C_D		"					"
	"	95							"							"	C_A		"					"
C_{IS}	3012	96	10/			10/	GND	GND	GND	10/			10/	GND	GND	5.0 V	I_A		20					pF
	"	97					"	"	"					"	"	"	I_B		"					"
	"	98					"	"	"					"	"	"	I_C		"					"
	"	99					"	"	"					"	"	"	I_D		"					"
C_{OS}		100		11/	11/		"	"	"		11/	11/		"	"	"	O_A		"					"
		101					"	"	"					"	"	"	O_B		"					"
		102					"	"	"					"	"	"	O_C		"					"
		103					"	"	"					"	"	"	O_D		"					"
C_{IOS}		104	12/	12/	12/	12/	"	"	"	12/	12/	12/	12/	"	"	"	I_A,O_A		"					"
		105					"	"	"					"	"	"	I_B,O_B		"					"
		106					"	"	"					"	"	"	I_C,O_C		"					"
		107					"	"	"					"	"	"	I_D,O_D		"					"
																		Subgroup 9 T_C = 25°C Min	Subgroup 9 Max	Subgroup 10 T_C = 125°C Min	Subgroup 10 Max	Subgroup 11 T_C = -55°C Min	Subgroup 11 Max	
t_{PLH}	3003	108	IN	OUT	OUT	IN	GND	GND	GND	IN	OUT	OUT	IN	GND	5.0 V	5.0 V	I_A to O_A	4	70	7	85	4	70	ns
	Fig. 5	109					5.0 V	"	"					"	GND	"	I_B to O_B	"	"	"	"	"	"	"
	"	110					GND	5.0 V	"					"	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	111					"	GND	"					5.0 V	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHL}	"	112	IN	OUT	OUT	IN	"	"	"	IN	OUT	OUT	IN	GND	5.0 V	"	I_A to O_A	"	60	"	80	"	50	"
	"	113					5.0 V	"	"					"	GND	"	I_B to O_B	"	"	"	"	"	"	"
	"	114					GND	5.0 V	"					"	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	115					"	GND	"					5.0 V	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHZ}	Fig. 6	116	5.0 V	OUT	OUT	5.0 V	"	"	"	5.0 V	OUT	OUT	5.0 V	GND	IN	"	C_A to O_A	"	70	"	90	"	55	"
	"	117					IN	IN	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	118					GND	"	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	119					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZH}	Fig. 6	120	5.0 V	OUT	OUT	5.0 V	"	"	"	5.0 V	OUT	OUT	5.0 V	GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	121					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	122					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	123					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PLZ}	Fig. 6	124	GND	OUT	OUT	5.0 V	"	"	"	GND	OUT	OUT	GND	GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	125					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	126					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	127					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZL}	Fig. 6	128	GND	OUT	OUT	5.0 V	"	"	"	GND	OUT	OUT	GND	GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	129					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	130					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	131					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"

1/ Pins not designated may be "High" level logic, "Low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open.
2/ I_{SS} measurements shall be run in sequence.
3/ When performing quiescent current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter. The outputs during the I_{SS} measurement shall be open.
4/ V_{IH1} = 3.8 V at 25°C, 3.6 V at 125°C, 3.95 V at -55°C.
V_{IH2} = 11.4 V at 25°C, 10.95 V at 125°C, 11.85 V at -55°C.
V_{IL1} = 1.1 V at 25°C, 0.85 V at 125°C, 1.25 V at -55°C.
V_{IL2} = 3.3 V at 25°C, 2.55 V at 125°C, 4.05 V at -55°C.
5/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
6/ See figure 7.
7/ See figure 8.
8/ See figure 9.
9/ See figure 10.
10/ C_C and C_{IS} - connect capacitance bridge between measured input terminal and V_{SS}, frequency = 1 MHz.
11/ C_{OS} - connect capacitance bridge between measured output terminal and V_{SS}, frequency = 1 MHz.
12/ C_{IOS} - connect capacitance bridge between measured input and output terminals, frequency = 1 MHz.

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TABLE III. Group A inspection for device type 51 - Continued.

Symbol	MIL-STD-883 method	Cases A, C,D,X,Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test limits						Unit
																		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1													1 mA	GND	C_A		1.5					V
		2					1 mA									"	C_B		"					"
		3						1 mA								"	C_C		"					"
		4												1 mA		"	C_D		"					"
$V_{IC(neg)}$		5													-1 mA		C_A		-6					"
		6					-1 mA										C_B		"					"
		7						-1 mA									C_C		"					"
		8												-1 mA			C_D		"					"
I_{SS} 2/ 3/	3005	9	18 V	GND	GND	18 V	GND	GND	GND	18 V	GND	GND	18 V	GND	GND	18 V	V_{SS}		-30		-550			nA
	"	10	GND	18 V	18 V	GND	"	"	"	GND	18 V	18 V	GND	"	"	"	"		"		"			"
	"	11	"			"	18 V	18 V	"	"	GND	GND	"	18 V	18 V	"			"		"			"
V_{OH3}	3006	12	"			15 V	15 V	GND	"	"			"	GND	GND	15 V	O_B	14.95		14.95		14.95		V
	"	13	"			GND	GND	15 V	"	"			"	"	"	"	O_C	"		"		"		"
	"	14	"			"	"	GND	"	15 V			15 V	15 V	GND	"	O_D	"		"		"		"
		15	15 V			"	"	"	"	GND			GND	GND	15 V	"	O_A	"		"		"		"
V_{OL3}	3007	16	GND			"	15 V	"	"	15 V			"	"	GND	5 V	O_B		.05		.05		.05	"
	"	17	"			15 V	GND	15 V	"	GND			"	"	"	"	O_C		"		"		"	"
	"	18	15 V			GND	"	GND	"	"			"	15 V	GND	"	O_D		"		"		"	"
		19	GND			"	"	"	"	"			15 V	GND	15 V	"	O_A		"		"		"	"
V_{IH1}		20	"			5.0 V	3.5 V	"	"	5.0 V			GND	"	GND	5 V	O_B	4.5		4.5		4.5		"
		21	"			GND	GND	3.5 V	"	GND			"	"	"	"	O_C	"		"		"		"
		22	"			"	"	GND	"	"			5.0 V	3.5 V	1.5 V	"	O_D	"		"		"		"
		23	5.0 V			"	"	"	"	"			GND	1.5 V	3.5 V	"	O_A	"		"		"		"
V_{IH2}		24	GND			10 V	7.0 V	"	"	"			"	GND	GND	10.0 V	O_B	9.0		9.0		9.0		"
		25	"			GND	GND	7.0 V	"	GND			"	7.0 V	3.0 V	"	O_C	"		"		"		"
		26	"			"	"	GND	"	"			10 V	7.0 V	3.0 V	"	O_D	"		"		"		"
		27	10 V			"	"	"	"	"			GND	3.0 V	7.0 V	"	O_A	"		"		"		"
V_{IH3}		28	GND			15 V	11.0 V	"	"	"			"	GND	GND	15 V	O_B	13.5		13.5		13.5		"
		29	"			GND	"	11.0 V	"	GND			"	11.0 V	4.0 V	"	O_C	"		"		"		"
		30	"			"	"	GND	"	"			15 V	4.0 V	11.0 V	"	O_D	"		"		"		"
		31	15 V			"	"	"	"	"			GND	4.0 V	11.0 V	"	O_A	"		"		"		"
V_{IL1}		32	GND			5.0 V	1.0 V	"	"	"			"	GND	GND	5.0 V	O_B		0.5		0.5		0.5	"
		33	"			GND	"	1.0 V	"	5.0 V			"	"	"	"	O_C		"		"		"	"
		34	"			"	"	GND	"	GND			5.0 V	1.0 V	GND	"	O_D		"		"		"	"
		35	5.0 V			"	"	"	"	"			GND	GND	1.0 V	"	O_A		"		"		"	"
V_{IL2}		36	GND			10.0 V	2.0 V	GND	"	"			"	"	GND	10.0 V	O_B		1.0		1.0		1.0	"
		37	"			GND	"	2.0 V	"	10.0 V			"	"	"	"	O_C		"		"		"	"
		38	"			"	"	GND	"	"			10.0 V	2.0 V	2.0 V	"	O_D		"		"		"	"
		39	10.0 V			"	"	"	"	"			GND	GND	2.0 V	"	O_A		"		"		"	"
V_{IL3}		40	GND			15.0 V	2.5 V	"	"	"			"	"	GND	15.0 V	O_B		1.5		1.5		1.5	"
		41	"			GND	"	2.5 V	"	15.0 V			"	"	"	"	O_C		"		"		"	"
		42	"			"	"	GND	"	"			15.0 V	2.5 V	2.5 V	"	O_D		"		"		"	"
		43	15.0 V			"	"	"	"	"			GND	GND	2.5 V	"	O_A		"		"		"	"
I_{OL1}		44	GND		0.4 V	"	5.0 V	"	"	5.0 V	0.4 V		"	"	GND	5 V	O_B	0.51		0.36		0.64		mA
		45	"			5.0 V	GND	5.0 V	"	GND		0.4 V	"	"	"	"	O_C	"		"		"		"
		46	5.0 V	0.4 V		GND	"	GND	"	"			"	5.0 V	5.0 V	"	O_D	"		"		"		"
		47	GND			"	"	"	"	"			5.0 V	GND	"	"	O_A	"		"		"		"
I_{OL2}		48	"		1.5 V	"	15 V	"	"	15 V	1.5 V		GND	"	GND	15 V	O_B	3.4		2.4		4.2		"
		49	"			15 V	GND	15 V	"	GND		1.5 V	"	"	"	"	O_C	"		"		"		"
		50	15 V	1.5 V		GND	"	GND	"	"			"	15 V	15 V	"	O_D	"		"		"		"
		51	GND			"	"	"	"	"			15 V	GND	"	"	O_A	"		"		"		"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 51 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits Subgroup 1 T_C = 25°C Min	Subgroup 1 T_C = 25°C Max	Subgroup 2 T_C = 125°C Min	Subgroup 2 T_C = 125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
I_{OH1}		52	GND		4.6 V	5.0 V	5.0 V	GND	GND	GND	4.6 V		GND	GND	GND	5.0 V	O_B	-0.51		-0.36		-0.64		mA
		53	"			GND	GND	5.0 V	"	5.0 V		4.6 V	"	"	"	"	O_C	"		"		"		"
		54	"			"	"	GND	"	GND			5.0 V	5.0 V	"	"	O_D	"		"		"		"
		55	5.0 V	4.6 V		"	"	"	"	"			GND	GND	5.0 V	"	O_A	"		"		"		"
I_{OH2}		56	GND		13.5 V	15 V	15 V	"	"	"	3.5 V		"	"	"	15 V	O_B	-3.4		-2.4		-4.2		"
		57	"			GND	GND	15 V	"	15 V		13.5 V	"	"	"	"	O_C	"		"		"		"
		58	"			"	"	GND	"	GND			15 V	15 V	"	"	O_D	"		"		"		"
		59	15 V	13.5 V		"	"	"	"	"			GND	GND	15 V	"	O_A	"		"		"		"
I_{IH1} 4/	3010	60	18 V			18 V	18 V	18 V	"	18 V			10 V	10 V	10 V	18 V	All inputs together		+8					nA
I_{IH2}	"	61	GND			18 V	GND	GND	"	GND			GND	GND	GND	"	C_B		1.0		45			"
	"	62	"			GND	"	18 V	"	"			"	"	"	"	C_C		"		"			"
	"	63	"			"	"	GND	"	"			"	18 V	"	"	C_D		"		"			"
	"	64	"			"	"	"	"	"			"	GND	18 V	"	C_A		"		"			"
	"	65	18 V			"	"	"	"	"			"	"	GND	"	I_A		"		"			"
	"	66	GND			18 V	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	67	"			GND	"	"	"	18 V			"	"	"	"	I_C		"		"			"
	"	68	"			"	"	"	"	GND			18 V	"	"	"	I_D		"		"			"
I_{IL1} 4/	3009	69	"			"	"	"	"	"			"	"	"	"	All inputs together		-8.0					"
I_{IL2}	"	70	GND			GND	GND	GND	GND	GND			GND	GND	18 V	GND	C_B		-1.0		-45			"
	"	71	"			"	"	"	"	"			"	"	"	"	C_C		"		"			"
	"	72	"			"	"	"	"	"			"	"	"	"	C_D		"		"			"
	"	73	"			"	"	"	"	"			"	"	"	"	C_A		"		"			"
	"	74	"			"	"	"	"	"			"	"	"	"	I_A		"		"			"
	"	75	"			"	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	76	"			"	"	"	"	"			"	"	"	"	I_C		"		"			"
	"	77	"			"	"	"	"	"			"	"	"	"	I_D		"		"			"
R_{ON1}		78	0.5 V	5/											5.0 V	5.0 V	O_A		700		750		550	Ω
		79			5/	0.5 V	5.0 V										O_B		"		"		"	"
		80						5.0 V		0.5 V	5/						O_C		"		"		"	"
		81										5/	0.5 V	5.0 V			O_D		"		"		"	"
		82	1.0 V	5/											5.0 V		O_A		1.0 k		1.1 k		750	"
		83			5/	1.0 V	5.0 V										O_B		"		"		"	"
		84						5.0 V		1.0 V	5/						O_C		"		"		"	"
		85										5/	1.0 V	5.0 V			O_D		"		"		"	"
		86	4.0 V	5/											5.0 V		O_A		1.2 k		1.8 k		850	"
		87			5/	4.0 V	5.0 V										O_B		"		"		"	"
		88						5.0 V		4.0 V	5/						O_C		"		"		"	"
		89										5/	4.0 V	5.0 V			O_D		"		"		"	"
		90	5.0 V	5/											5.0 V		O_A		800		950		700	"
		91			5/	5.0 V	5.0 V										O_B		"		"		"	"
		92						5.0 V		5.0 V	5/						O_C		"		"		"	"
		93										5/	5.0 V	5.0 V			O_D		"		"		"	"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 51 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits						Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
																		Min	Max	Min	Max	Min	Max	
R_{ON2}		94	-5.0 V	6/					-7.5 V						7.5 V	7.5 V	O_A		250		300		250	Ω
		95			6/	-5.0 V	7.5 V		"							"	O_B		"		"		"	"
		96						7.5 V	"	-5.0 V	6/					"	O_C		"		"		"	"
		97							"			6/	-5.0 V	7.5 V		"	O_D		"		"		"	"
		98	0.6 V	6/					"						7.5 V	"	O_A		400		450		300	"
		99			6/	0.6 V	7.5 V		"							"	O_B		"		"		"	"
		100						7.5 V	"	0.6 V	6/					"	O_C		"		"		"	"
		101							"			6/	0.6 V	7.5 V		"	O_D		"		"		"	"
		102	5.0 V	6/					"						7.5 V	"	O_A		350		400		"	"
		103			6/	5.0 V	7.5 V		"							"	O_B		"		"		"	"
		104						7.5 V	"	5.0 V	6/					"	O_C		"		"		"	"
		105							"			6/	5.0 V	7.5 V		"	O_D		"		"		"	"
R_{ON3}		106	2.5 V	7/					GND						10 V	10 V	O_A		450		500		350	"
		107			7/	2.5 V	10 V		"							"	O_B		"		"		"	"
		108						10 V	"	2.5 V	7/					"	O_C		"		"		"	"
		109							"			7/	2.5 V	10 V		"	O_D		"		"		"	"
		110	5.0 V	7/					"						10 V	"	O_A		650		700		500	"
		111			7/	5.0 V	10 V		"							"	O_B		"		"		"	"
		112						10 V	"	5.0 V	7/					"	O_C		"		"		"	"
		113							"			7/	5.0 V	10 V		"	O_D		"		"		"	"
		114	7.5 V	7/					"						10 V	"	O_A		600		750		"	"
		115			7/	7.5 V	10 V		"							"	O_B		"		"		"	"
		116						10 V	"	7.5 V	7/					"	O_C		"		"		"	"
		117							"			7/	7.5 V	10 V		"	O_D		"		"		"	"
I_{DOFF1}		118	8/	8/	8/	8/	8/	8/	GND	8/	8/	8/	8/	8/	8/	5.0 V	8/		45		45		45	nA
I_{DOFF2}		119	"	"	"	"	"	"	-5.0 V	"	"	"	"	"	"	GND	"		"		"		"	"
I_{DOFF3}		120	"	"	"	"	"	"	GND	"	"	"	"	"	"	10 V	"		"		"		"	"
I_{DOFF4}		121	"	"	"	"	"	"	-10.0 V	"	"	"	"	"	"	GND	"		"		"		"	"
I_{DOFF5}		122	"	"	"	"	"	"	GND	"	"	"	"	"	"	15 V	"		"		"		"	"
I_{DOFF6}		123	"	"	"	"	"	"	-15.0 V	"	"	"	"	"	"	GND	"		"		"		"	"
																		Subgroup 4 T_C = 25°C						
																		Min	Max					
C_C	3012	124					9/	9/	GND					9/		GND	C_B		12					pF
	"	125							"							"	C_C		"					"
	"	126							"							"	C_D		"					"
	"	127							"						9/	"	C_A		"					"
C_{is}	"	128	9/						"							"	I_A		15					"
	"	129				9/	GND	GND	"					GND	GND	"	I_B		"					"
	"	130					"	"	"	9/				"	"	"	I_C		"					"
	"	131							"				9/			5.0 V	I_D		"					"
C_{os}	"	132		10/					"							"	O_A		15					"
	"	133			10/		"	"	"					"	"	"	O_B		"					"
	"	134							"		10/					"	O_C		"					"
	"	135							"			10/				"	O_D		"					"
C_{ios}	"	136	11/	11/					"							"	I_A, O_A		15					"
	"	137			11/	11/	"	"	"					"	"	"	I_B, O_B		"					"
	"	138							"	11/	11/					"	I_C, O_C		"					"
	"	139							"			11/	11/			"	I_D, O_D		"					"

See footnotes at end of device type 52

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TABLE III. Group A inspection for device type 51 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C,D,X,Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits						Unit
																		Subgroup 9 T_C = 25°C		Subgroup 10 T_C = 125°C		Subgroup 11 T_C = -55°C		
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	$-C_A$	V_{DD}		Min	Max	Min	Max	Min	Max	
t_{PLH}	3003 Fig. 5	140	*IN*	*OUT*			*GND*	*GND*	*GND*					*GND*	*5.0 V*	*5.0 V*	I_A to O_A	*4*	*125*	*7*	*130*	*4*	*114*	*ns*
	"	141			OUT	IN	5.0 V	"	"					"	GND	"	I_B to O_B	"	"	"	"	"	"	"
	"	142					GND	5.0 V	"	IN	OUT			"	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	143					"	GND	"			OUT	IN	5.0 V	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHL}	"	144	IN	OUT			GND	GND	GND					GND	5.0 V	"	I_A to O_A	"	130	"	156	"	130	"
	"	145			OUT	IN	5.0 V	"	"					"	GND	"	I_B to O_B	"	"	"	"	"	"	"
	"	146					GND	5.0 V	"	IN	OUT			"	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	147					"	5.0 V	"			OUT	IN	5.0 V	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHZ}	Fig. 6	148	5.0 V	OUT			"	"	"					GND	IN	"	C_A to O_A	"	70	"	90	"	50	"
	"	149			OUT	5.0 V	IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	150					GND	IN	"	5.0 V	OUT			"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	151					"	GND	"			OUT	5.0 V	IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZH}	"	152	5.0 V	OUT			"	"	"					GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	153			OUT	5.0 V	IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	154					GND	IN	"	5.0 V	OUT			"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	155					"	GND	"			OUT	5.0 V	IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PLZ}	"	156	GND	OUT			"	"	"					GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	157			OUT	GND	IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	158					GND	IN	"	GND	OUT			"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	159					"	GND	"			OUT	GND	IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZL}	"	160	GND	OUT			"	"	"					GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	161			OUT	GND	IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	162					GND	IN	"	GND	OUT			"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	163					"	GND	"			OUT	GND	IN	"	"	C_D to O_D	"	"	"	"	"	"	"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits						Unit
																		Subgroup 1 T_C = 25°C		Subgroup 2 T_C = 125°C		Subgroup 3 T_C = -55°C		
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}		Min	Max	Min	Max	Min	Max	
$V_{IC(pos)}$		1					1 mA									GND	O_B		1.5					V
		2						1 mA						1 mA		"	C_C		"					"
		3													1 mA	"	C_D		"					"
		4														"	C_A		"					"
$V_{IC(neg)}$		5					-1 mA		GND							"	O_B		-6					"
		6						-1 mA	"					-1 mA		"	O_C		"					"
		7							"						-1 mA	"	O_D		"					"
		8							"							"	O_A		"					"
I_{SS} 2/3/	3005	9	18 V	GND	GND	18 V	GND	GND	"	18 V	GND	GND	18 V	GND	GND	18 V	V_{SS}		-30		-550			nA
	"	10	GND	18 V	18 V	GND	18 V	"	"	GND	18 V	18 V	GND	18 V	18 V	"	"		"		"			"
	"	11	"	"	"	"	"	18 V	"	"	"	"	"	"	"	"	"		"		"			"
V_{OH3}	3006	12	"			15 V	15 V	GND	"	"			"	GND	GND	15 V	O_B	14.95		14.95		14.95		V
	"	13	"			GND	GND	15 V	"	15 V			15 V	15 V	"	"	O_C	"		"		"		"
	"	14	"			"	"	GND	"	GND			GND	GND	15 V	"	O_D	"		"		"		"
	"	15	15 V			"	"	"	"	"			GND	GND	"	"	O_A	"		"		"		"
V_{OL3}	3007	16	GND			"	15 V	"	"	15 V			"	"	GND	"	O_B		.05		.05		.05	"
	"	17	"			15 V	GND	15 V	"	GND			"	15 V	"	"	O_C		"		"		"	"
	"	18	15 V			GND	"	GND	"	"			"	GND	15 V	"	O_D		"		"		"	"
	"	19	GND			"	"	"	"	"			15 V	"	"	"	O_A		"		"		"	"
V_{IH1}		20	"			5.0 V	3.5 V	GND	"	"			GND	GND	GND	5.0 V	O_B	4.5		4.5		4.5		"
		21	"			GND	GND	3.5 V	"	5.0 V			5.0 V	3.5 V	"	"	O_C	"		"		"		"
		22	"			"	"	GND	"	GND			GND	GND	3.5 V	"	O_D	"		"		"		"
		23	5.0 V			"	"	"	"	"			"	"	"	"	O_A	"		"		"		"
V_{IH2}		24	GND			10 V	7.0 V	"	"	"			"	GND	GND	10.0 V	O_B	9.0		9.0		9.0		"
		25	"			GND	GND	7.0 V	"	10 V			10 V	7.0 V	"	"	O_C	"		"		"		"
		26	"			"	"	GND	"	GND			GND	GND	7.0 V	"	O_D	"		"		"		"
		27	10.0 V			"	"	"	"	"			"	"	"	"	O_A	"		"		"		"
V_{IH3}		28	GND			15 V	11.0 V	"	"	"			"	GND	GND	15 V	O_B	13.5		13.5		13.5		"
		29	"			GND	GND	11.0 V	"	15.0 V			15 V	11.0 V	"	"	O_C	"		"		"		"
		30	"			"	"	GND	"	GND			GND	GND	11.0 V	"	O_D	"		"		"		"
		31	15 V			"	"	"	"	"			"	"	"	"	O_A	"		"		"		"
V_{IL1}		32	GND			5.0 V	1.5 V	GND	"	"			GND	GND	GND	5.0 V	O_B		0.5		0.5		0.5	"
		33	"			GND	GND	1.5 V	"	5.0 V			5.0 V	1.5 V	"	"	O_C		"		"		"	"
		34	"			"	"	GND	"	GND			GND	GND	1.5 V	"	O_D		"		"		"	"
		35	5.0 V			"	"	"	"	"			"	"	"	"	O_A		"		"		"	"
V_{IL2}		36	GND			10.0 V	3.0 V	"	"	"			"	"	GND	10.0 V	O_B		1.0		1.0		1.0	"
		37	"			GND	GND	3.0 V	"	10.0 V			10.0 V	3.0 V	"	"	O_C		"		"		"	"
		38	"			"	"	GND	"	GND			GND	GND	3.0 V	"	O_D		"		"		"	"
		39	10.0 V			"	"	"	"	"			"	"	"	"	O_A		"		"		"	"
V_{IL3}		40	GND			15.0 V	4.0 V	"	"	"			"	GND	GND	15.0 V	O_B		1.5		1.5		1.5	"
		41	"			GND	GND	4.0 V	"	15.0 V			15.0 V	4.0 V	"	"	O_C		"		"		"	"
		42	"			"	"	GND	"	GND			GND	GND	4.0 V	"	O_D		"		"		"	"
		43	15.0 V			"	"	"	"	"			"	"	"	"	O_A		"		"		"	"
I_{OL1}		44	GND		0.4 V	"	5.0 V	"	"	5.0 V	0.4 V		"	"	GND	5.0 V	O_B	0.51		0.36		0.64		mA
		45	"			5.0 V	GND	5.0 V	"	GND		0.4 V	"	5.0 V	"	"	O_C	"		"		"		"
		46	15.0 V	0.4 V		GND	"	GND	"	"			"	GND	5.0 V	"	O_D	"		"		"		"
		47	GND			"	"	"	"	"			5.0 V	"	"	"	O_A	"		"		"		"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits Subgroup 1 T_C = 25°C Min	Subgroup 1 T_C = 25°C Max	Subgroup 2 T_C = 125°C Min	Subgroup 2 T_C = 125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
I_{OL2}		48	GND		1.5 V	GND	15 V	GND	GND	15 V	1.5 V		GND	GND	GND	15 V	O_B	3.4		2.4		4.2		mA
		49	"			15 V	GND	15 V	"	GND			"		"	"	O_C	"		"		"		"
		50	15 V			GND	"	GND	"	"		1.5 V	"	15 V	"	"	O_D	"		"		"		"
		51	GND	1.5 V		"	"	"	"	"			15 V	GND	15 V	"	O_A	"		"		"		"
I_{OH1}		52	"		4.6 V	5.0 V	5.0 V	"	"	"	4.6 V		GND	"	GND	5.0 V	O_B	-0.51		-0.36		-0.64		"
		53	"			GND	GND	5.0 V	"	5.0 V			"	"	"	"	O_C	"		"		"		"
		54	"			"	"	GND	"	GND		4.6 V	5.0 V	5.0 V	"	"	O_D	"		"		"		"
		55	5.0 V	4.6 V		"	"	"	"	"			GND	GND	5.0 V	"	O_A	"		"		"		"
I_{OH2}		56	GND		13.5 V	15 V	15 V	"	"	"	13.5 V		"	"	GND	15 V	O_B	-3.4		-2.4		-4.2		"
		57	"			GND	GND	15 V	"	15 V			"	"	"	"	O_C	"		"		"		"
		58	"			"	"	GND	"	GND		13.5 V	15 V	15 V	"	"	O_D	"		"		"		"
		59	15 V	13.5 V		"	"	"	"	"			GND	GND	15 V	"	O_A	"		"		"		"
I_{IH1} 4/	3010	60	18 V			18 V	18 V	18 V	"	18 V			18 V	18 V	18 V	18 V	All inputs together		+8					nA
I_{IH2}	"	61	GND			GND	18 V	GND	"	GND			GND	GND	GND	"	C_B		1.0		45			"
	"	62	"			"	GND	18 V	"	"			"	"	"	"	C_C		"		"			"
	"	63	"			"	"	GND	"	"			"	18 V	"	"	C_D		"		"			"
	"	64	"			"	"	"	"	"			"	GND	18 V	"	C_A		"		"			"
	"	65	18 V			"	"	"	"	"			"	"	GND	"	I_A		"		"			"
	"	66	GND			18 V	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	67	"			GND	"	"	"	18 V			"	"	"	"	I_C		"		"			"
	"	68	"			"	"	"	"	GND			18 V	"	"	"	I_D		"		"			"
I_{IL1} 4/	3009	69	"			"	"	"	"	"			GND	"	"	"	All inputs together		-8.0					"
I_{IL2}	"	70	GND			GND	GND	GND	GND	GND			GND	GND	GND	GND	C_B		-1.0		-45			"
	"	71	"			"	"	"	"	"			"	"	"	"	C_C		"		"			"
	"	72	"			"	"	"	"	"			"	"	"	"	C_D		"		"			"
	"	73	"			"	"	"	"	"			"	"	"	"	C_A		"		"			"
	"	74	"			"	"	"	"	"			"	"	"	"	I_A		"		"			"
	"	75	"			"	"	"	"	"			"	"	"	"	I_B		"		"			"
	"	76	"			"	"	"	"	"			"	"	"	"	I_C		"		"			"
	"	77	"			"	"	"	"	"			"	"	"	"	I_D		"		"			"
R_{ON1}		78	0.5 V	5/					"						5.0 V	5.0 V	O_A		400		450		400	Ω
		79			5/	0.5 V	5.0 V		"							"	O_B		"		"		"	"
		80						5.0 V	"	0.5 V	5/					"	O_C		"		"		"	"
		81							"			5/	0.5 V	5.0 V		"	O_D		"		"		"	"
		82	1.0 V	5/					"						5.0 V	"	O_A		450		500		450	"
		83			5/	1.0 V	5.0 V		"							"	O_B		"		"		"	"
		84						5.0 V	"	1.0 V	5/					"	O_C		"		"		"	"
		85							"			5/	1.0 V	5.0 V		"	O_D		"		"		"	"
		86	4.0 V	5/					"						5.0 V	"	O_A		550		600		550	"
		87			5/	4.0 V	5.0 V		"							"	O_B		"		"		"	"
		88						5.0 V	"	4.0 V	5/					"	O_C		"		"		"	"
		89							"			5/	4.0 V	5.0 V		"	O_D		"		"		"	"
		90	5.0 V	5/					"						5.0 V	"	O_A		400		450		400	"
		91			5/	5.0 V	5.0 V		"							"	O_B		"		"		"	"
		92						5.0 V	"	5.0 V	5/					"	O_C		"		"		"	"
		93							"			5/	5.0 V	5.0 V		"	O_D		"		"		"	"

See footnotes at end of device type 52.

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TABLE III. Group A inspection for device type 52 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C, D, X, Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits Subgroup 1 T_C = 25°C Min	Subgroup 1 T_C = 25°C Max	Subgroup 2 T_C = 125°C Min	Subgroup 2 T_C = 125°C Max	Subgroup 3 T_C = -55°C Min	Subgroup 3 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
R_{ON2}		94	-5.0 V	6/					-7.5 V						7.5 V	7.5 V	O_A		200		250		200	Ω
		95			6/	-5.0 V	7.5 V		"							"	O_B		"		"		"	"
		96						7.5 V	"	-5.0 V	6/					"	O_C		"		"		"	"
		97							"			6/	-5.0 V	7.5 V		"	O_D		"		"		"	"
		98	0.6 V	6/					"						7.5 V	"	O_A		"		"		"	"
		99			6/	0.6 V	7.5 V		"							"	O_B		"		"		"	"
		100						7.5 V	"	0.6 V	6/					"	O_C		"		"		"	"
		101							"			6/	0.6 V	7.5 V		"	O_D		"		"		"	"
		102	5.0 V	6/					"						7.5 V	"	O_A		"		"		"	"
		103			6/	5.0 V	7.5 V		"							"	O_B		"		"		"	"
		104						7.5 V	"	5.0 V	6/					"	O_C		"		"		"	"
		105							"			6/	5.0 V	7.5 V		"	O_D		"		"		"	"
R_{ON3}		106	2.5 V	7/					GND						10 V	10 V	O_A		300		350		300	"
		107			7/	2.5 V	10 V		"							"	O_B		"		"		"	"
		108						10 V	"	2.5 V	7/					"	O_C		"		"		"	"
		109							"			7/	2.5 V	10 V		"	O_D		"		"		"	"
		110	5.0 V	7/					"						10 V	"	O_A		250		300		250	"
		111			7/	5.0 V	10 V		"							"	O_B		"		"		"	"
		112						10 V	"	5.0 V	7/					"	O_C		"		"		"	"
		113							"			7/	5.0 V	10 V		"	O_D		"		"		"	"
		114	7.5 V	7/					"						10 V	"	O_A		"		"		"	"
		115			7/	7.5 V	10 V		"							"	O_B		"		"		"	"
		116						10 V	"	7.5 V	7/					"	O_C		"		"		"	"
		117							"			7/	7.5 V	10 V		"	O_D		"		"		"	"
I_{DOFF1}	Fig. 10	118	8/	8/	8/	8/	8/	8/	GND	8/	8/	8/	8/	8/	8/	5.0 V	8/		-45		-45		-45	nA
I_{DOFF2}		119	"	"	"	"	"	"	-5.0 V	"	"	"	"	"	"	GND	"		45		45		45	"
I_{DOFF3}		120	"	"	"	"	"	"	GND	"	"	"	"	"	"	10 V	"		-45		-45		-45	"
I_{DOFF4}		121	"	"	"	"	"	"	-10.0 V	"	"	"	"	"	"	GND	"		45		45		45	"
I_{DOFF5}		122	"	"	"	"	"	"	GND	"	"	"	"	"	"	15 V	"		-45		-45		-45	"
I_{DOFF6}		123	"	"	"	"	"	"	-15.0 V	"	"	"	"	"	"	GND	"		45		45		45	"

																		Subgroup 4 T_C = 25°C Min	Subgroup 4 T_C = 25°C Max					
C_C	3012	124					9/	9/	GND					9/		GND	C_B		12					pF
	"	125							"							"	C_C		"					"
	"	126							"							"	C_D		"					"
	"	127							"						9/	"	C_A		"					"
C_{IS}	"	128	9/				GND	GND	"					GND	GND	5.0 V	I_A		20					"
	"	129				9/	"	"	"					"	"	"	I_B		"					"
	"	130					"	"	"	9/				"	"	"	I_C		"					"
	"	131							"				9/				I_D		"					"
C_{OS}	"	132		10/			"	"	"					"	"	"	O_A		"					"
	"	133			10/		"	"	"					"	"	"	O_B		"					"
	"	134					"	"	"		10/			"	"	"	O_C		"					"
	"	135							"			10/					O_D		"					"
C_{IOS}	"	136	11/	11/			"	"	"					"	"	"	I_A, O_A		"					"
	"	137			11/	11/	"	"	"					"	"	"	I_B, O_B		"					"
	"	138					"	"	"	11/	11/			"	"	"	I_C, O_C		"					"
	"	139							"			11/	11/				I_D, O_D		"					"

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TABLE III. Group A inspection for device type 52 - Continued. 1/

Symbol	MIL-STD-883 method	Cases A, C,D,X,Y	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Test Limits: Subgroup 9 T_C = 25°C Min	Subgroup 9 T_C = 25°C Max	Subgroup 10 T_C = 125°C Min	Subgroup 10 T_C = 125°C Max	Subgroup 11 T_C = -55°C Min	Subgroup 11 T_C = -55°C Max	Unit
		Test no.	I_A	O_A	O_B	I_B	C_B	C_C	V_{SS}	I_C	O_C	O_D	I_D	C_D	C_A	V_{DD}								
t_{PLH}	3003 Fig. 5	140	IN	OUT	OUT	IN	GND	GND	GND	IN	OUT	OUT	IN	GND	5.0 V	5.0 V	I_A to O_A	4	70	7	85	4	70	ns
	"	141					5.0 V	"	"					"	GND	"	I_B to O_B	"	"	"	"	"	"	"
	"	142					GND	5.0 V	"					"	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	143					"	GND	"					5.0 V	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHL}	"	144	IN	OUT	OUT	IN	GND	GND	GND	IN	OUT	OUT	IN	GND	5.0 V	"	I_A to O_A	"	60	"	80	"	50	"
	"	145					5.0 V	"	"					"	GND	"	I_B to O_B	"	"	"	"	"	"	"
	"	146					GND	5.0 V	"					"	"	"	I_C to O_C	"	"	"	"	"	"	"
	"	147					"	5.0 V	"					5.0 V	"	"	I_D to O_D	"	"	"	"	"	"	"
t_{PHZ}	Fig. 6	148	5.0 V	OUT	OUT	5.0 V	"	"	"	5.0 V	OUT	OUT	5.0 V	GND	IN	"	C_A to O_A	"	70	"	90	"	55	"
	"	149					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	150					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	151					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZH}	Fig. 6	152	5.0 V	OUT	OUT	5.0 V	"	"	"	5.0 V	OUT	OUT	5.0 V	GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	153					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	154					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	155					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PLZ}	Fig. 6	156	GND	OUT	OUT	GND	"	"	"	GND	OUT	OUT	GND	GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	157					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	158					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	159					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"
t_{PZL}	Fig. 6	160	GND	OUT	OUT	GND	"	"	"	GND	OUT	OUT	GND	GND	IN	"	C_A to O_A	"	"	"	"	"	"	"
	"	161					IN	"	"					"	GND	"	C_B to O_B	"	"	"	"	"	"	"
	"	162					GND	IN	"					"	"	"	C_C to O_C	"	"	"	"	"	"	"
	"	163					"	GND	"					IN	"	"	C_D to O_D	"	"	"	"	"	"	"

1/ Pins not designated may be "High" level logic, "Low" level logic, or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open.
2/ I_{SS} measurements shall be run in sequence.
3/ When performing quiescent current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter. The outputs during the I_{SS} measurement shall be open.
4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.
5/ See figure 7.
6/ See figure 8.
7/ See figure 9.
8/ See figure 10.
9/ C_C and C_{is} - connect capacitance bridge between measured input terminal and V_{SS}, frequency = 1 MHz.
10/ C_{os} - connect capacitance bridge between measured output terminal and V_{SS}, frequency = 1 MHz.
11/ C_{ios} - connect capacitance bridge between measured input and output terminals, frequency = 1 MHz.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 4, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and shall be performed on each inspection lot for initially qualified device types 01 and 02, and measured only for initial qualification and after process or design changes for initially qualified device types 51 and 52. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 and as shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein. For class S devices, the total dose radiation testing shall be included as an additional test in the wafer lot acceptance of method 5007 of MIL-STD-883.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cooldown procedures. Burn-in and life test cooldown procedures shall be in accordance with MIL-STD-883 methods 1005 and 1015.

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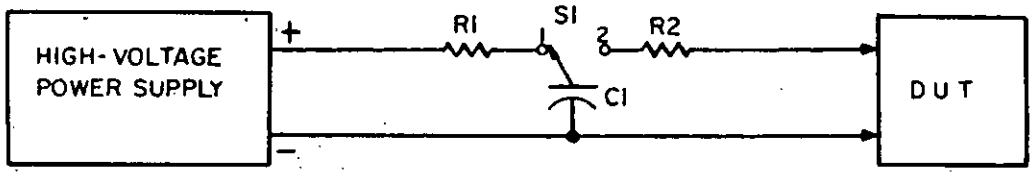
TABLE IV. Delta limits @25°C.

Parameter 1/	Device types	
	01,02	51, 52
I_{SS}	±25 nA	±25 nA
V_{OL1}	±.04 V	---
V_{OH1}	±.08 V	---
I_{OL1}	---	±15%
I_{OH1}	---	±15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

4.5.3 High voltage (V_{ZAP}) test or input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 11.

a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.



V_{ZAP} = 400 V charge on C1.

1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay

FIGURE 11. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Control input
2	Control input	V_{SS}

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c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS} test). When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 12. In situ and remote testing shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}	
		Device types 01, 02	Device types 51, 52
V_{TN}	0.3 V min	V_{DD} = 10 V	V_{DD} = 10 V
V_{TP}	2.8 V max	V_{DD} = 10 V	V_{DD} = 10 V
ΔV_T	1.4 V	V_{DD} = 10 V	V_{DD} = 10 V
I_{SS}	100 x max limit	V_{DD} = 15 V	V_{DD} = 18 V
t_{PLH}	1.35 x max limit	V_{DD} = 5 V	V_{DD} = 5 V
t_{PHL}	1.35 x max limit	V_{DD} = 5 V	V_{DD} = 5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 V dc (through a 30- to 60 kΩ resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01, 51	1, 4, 5, 6, 8, 11, 12, 13	7	14
02, 52	1, 4, 5, 6, 8, 11, 12, 13	7	14

Pins not designated are open or connected to 10 V dc through a 30- to 60 kilohm resistor.

4.6 Data reporting. When specified in the acquisition documnent, a copy of the following data, as applicable, shall be supplied.

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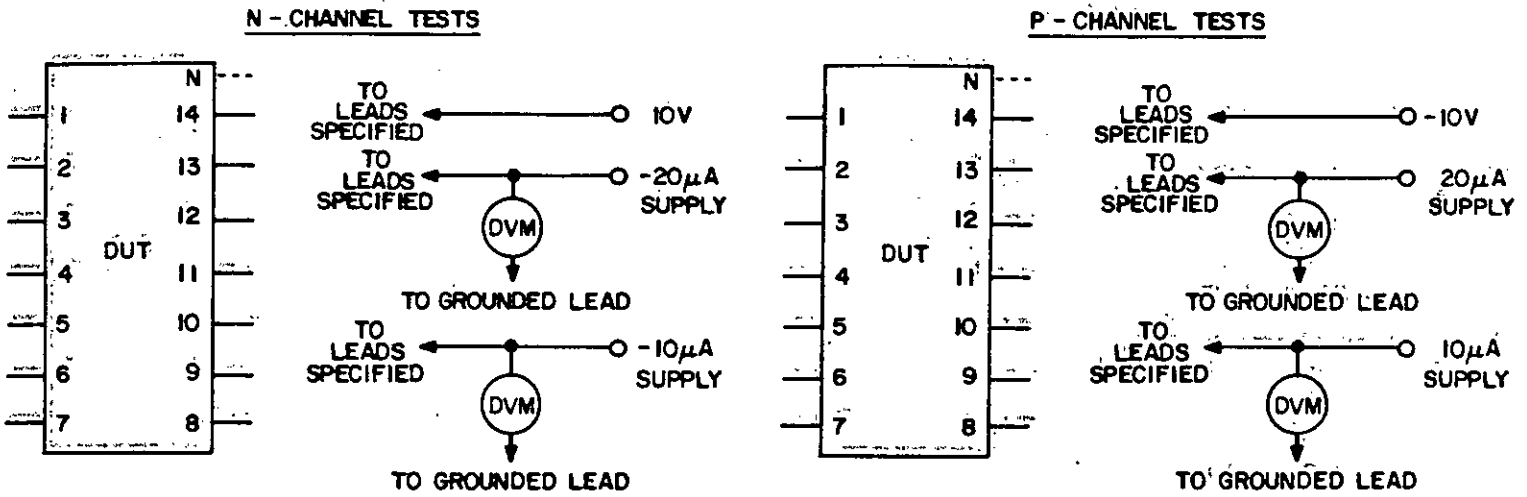


FIGURE 12. Threshold-voltage test circuit.

TABLE VIII. Threshold - voltage test circuit conditions.

Device type	GND	10 V	V_{TN} measured at		GND	10 V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01, 51	13	5, 6, 12, 14		7	13	5-7, 12		14
02, 52	13	5, 6, 12, 14		7	13	5-7, 12		14

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a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and steady state life tests (see 3.5).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parmeters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_I and C_C	Input terminal-to-V_{SS} capacitance.
C_{ios}	Switch input to output capacitance.
C_{os}	Output terminal-to-V_{SS} capacitance.
GND	Ground. Zero voltage potential.
I_{SS}	Quiescent supply current.
R_{on}	Switch on resistance in voltage using 1 mA current.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.
V_{is}	Input signal.
V_{os}	Output signal.
V_{SS}	Negative supply voltage.
V_{ZAP}	Input test voltage.

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6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	4016A
02	4066A
51	4016B
52	4066B

6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM, AR
Navy - AS, CG, MC, OS, SH

Preparing activity:
NASA - NA

Agent:
DLA - ES

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MIL-M-38510/58C	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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